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A Wobbling Ratio for diagnosing phase evolution of the Ulleung Warm Eddy from its three-dimensional tilt structure

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Introduction: The Ulleung Warm Eddy (UWE), a quasi-stationary mesoscale eddy in the East Sea, exhibits distinct phases over its lifespan. Identifying these phases is essential for understanding the evolution of UWE structure and associated energy transfer. Although the energy-budget framework has been widely used for eddy phase analysis, its interpretation is constrained by the energy-balance closure problem. To address this limitation, we proposed the Wobbling Ratio (WR), a quantitative index derived from the three-dimensional (3D) tilt structure of the UWE.

Methods: We produced subsurface temperature (Sub_T) and salinity (Sub_S) fields from surface data using a conditional computation model (CCM). Using the reconstructed 3D structures, we compared the energy budget and WR in relation to eddy volume.

Results: In the subset of the independent test set within the Ulleung Basin, RMSE remained below 1.62 °C and 0.07 psu across all depths, indicating that the CCM provided a stable basis for subsequent 3D eddy analysis. The reconstructed 3D structure of UWE15 showed a sequential morphological transition from an S-lens structure to a D-lens structure and eventually to a bowl-shaped structure. This morphological evolution was qualitatively consistent with the reported seasonal variability of East Korea Warm Current (EKWC) transport. WR showed a U-shaped evolution, decreasing from P1 to P2 in terms of mean WR (1.44 to 0.74) and then increasing through P3 and P4 (1.13 and 2.11, respectively). This pattern corresponded to a three-phase sequence in which the UWE transitioned from unstable growth to stable growth and finally to decay. Among the energy-budget diagnostics and WR, WR showed the largest absolute correlation with eddy volume, with a significant negative correlation ($r = -0.57$, $p = 1.03 \times 10^{-19}$).

Discussion: These results indicate that WR provides a quantitatively supported index for describing the volume-based phase evolution of the UWE.

KEYWORDS

deep learning, East Sea, eddy phase analysis, mesoscale eddy, subsurface reconstruction, three-dimensional eddy structure, Ulleung Warm Eddy, Wobbling Ratio

1 Introduction

The Ulleung Warm Eddy (UWE) is a mesoscale eddy in the Ulleung Basin of the East Sea, located between the Korean Peninsula and Japan (Figure 1A; Korea Hydrographic and Oceanographic Agency (KHOA), 2016). The UWE has an average lifetime of approximately eight months (Lee and Niiler, 2010; Shin et al., 2019), satisfies the long-lived eddy criterion (Chelton et al., 2011), and forms repeatedly at the same location. Recurrent formation makes the UWE a key factor in regional energy pathways, including energy transfer into the deep ocean (Holland, 1978; Noh and Nam, 2020, Noh and Nam, 2021). Accordingly, understanding the UWE lifespan is necessary to explain when and how this energy pathway operates in the East Sea.

The lifespan of mesoscale eddies is commonly partitioned into kinematic phases such as growth, maintenance, and decay (Valencia et al., 2025; Zhang et al., 2022). Accordingly, phase identification is a first step in understanding eddy lifespan. Transitions between phases involve changes in energy conversion and redistribution; therefore, quantitative diagnostics of the energy system are required (Guo et al., 2022; Tedesco et al., 2022). As a classical framework, Orlanski and Cox (1973) proposed a set of equations for the energy budget, which has been widely applied in eddy studies (Li et al., 2020; Oey, 2008; Storch et al., 2012).

This framework provides valuable dynamical insight into phase evolution of mesoscale eddy. However, its interpretation may be sensitive to the treatment of specific terms, such as nonlinear advection and vertical flux-related contributions (Arnaud et al., 2025; Evans et al., 2022; Wilder et al., 2023). These limitations do not invalidate energy-budget analysis, but they suggest that energy

diagnostics alone may not fully describe the kinematic phase of mesoscale eddy. In light of this, this study explores complementary approaches to energy-budget diagnostics, with a focus on the UWE.

An emerging approach for phase analysis is to use the three-dimensional (3D) tilt structure of mesoscale eddies (Han et al., 2025; Liu et al., 2024; Men et al., 2024). Such structural information, represented by eddy tilt and morphology, has been used to qualitatively assess external forcing and associated influences on eddy kinematic phases. For example, Cui et al. (2019) linked eddy splitting and merging to unstable multicore structures shaped by interactions with background currents, topography, and neighboring eddies; Li et al. (2020) used the 3D structure of a cyclonic eddy (CE)–anticyclonic eddy (AE) pair to distinguish life-cycle stages under background flow. Moreover, the 3D tilt structure provides information for interpreting nonlinear processes that are difficult to diagnose from energy-budget analyses alone (Li et al., 2023; Zhang et al., 2024).

Nevertheless, analyses of 3D tilt structure have remained largely at a qualitative level. Without quantitative measures, inferred physical processes remain descriptive and are difficult to validate rigorously (Tomczak, 2005). Progress therefore requires extending analyses of 3D tilt structure from qualitative description to quantitative diagnosis. Achieving this goal requires parameterizing the nonlinear relationship between physical variables and the 3D tilt structure into a single index (Eden and Greatbatch, 2008; Gent, 2011; Jansen et al., 2015); however, few studies have attempted to quantify this relationship.

In this study, we proposed the Wobbling Ratio (WR), a quantitative index derived from the 3D tilt structure of the UWE, and applied it to analyze UWE phases. For this, we first produced subsurface fields from surface data using a conditional computation

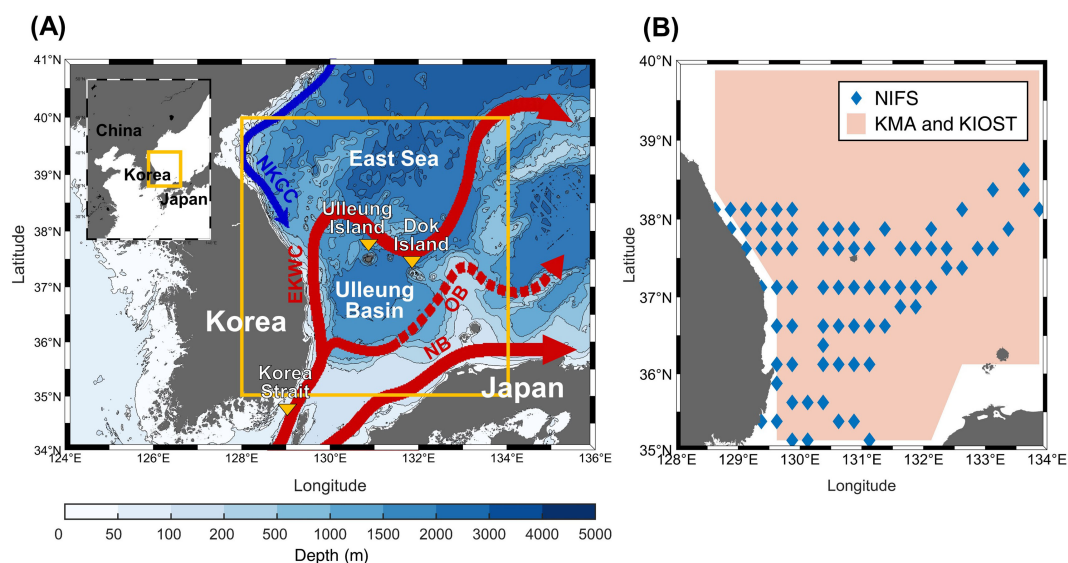


FIGURE 1

(A) Topography (shading) of the East Sea, located between the Korean Peninsula and Japan. The main study area is outlined by the yellow box. Yellow triangles denote the location of Ulleung Island, Dok Island, the Korea Strait. Major currents are shown schematically: red solid lines indicate the East Korea Warm Current (EKWC) and the nearshore branch (NB) of the Tsushima Warm Current (TWC); the red dashed line indicates the offshore branch (OB) of the TWC; and the blue line indicates the North Korea Cold Current (NKCC). The current was sourced from the Korea Hydrographic and Oceanography Agency (KHOA) (2016). (B) Spatial distribution of *in situ* observations collected within the main study area. Blue diamonds indicate stations from the National Institute of Fisheries Science (NIFS), and red shading indicates the spatial coverage of Argo observations provided by the Korea Institute of Ocean Science and Technology (KIOST) and the Korea Meteorological Administration (KMA).

model (CCM). The produced subsurface fields were then used to determine the UWE 3D structure, including tilt and morphology, through Lagrangian particle tracking. Finally, we applied WR to analyze the kinematic phases of the UWE.

2 Data

Sea surface and subsurface data were used as model inputs and outputs for training, respectively, consistent with steric and non-steric contributions to sea level (Gill and Niller, 1973) (Table 1). The data cover 1993–2023 over 35–40°N and 128–132°E. All data were processed using MATLAB (The MathWorks, Inc., 2023) and Python (Python Software Foundation, 2023).

2.1 Sea surface data

Sea surface height (SSH) was obtained from the Copernicus Marine Environment Monitoring Service (CMEMS) absolute dynamic topography (ADT) product (Pujol et al., 2016) at daily $0.25^\circ \times 0.25^\circ$ resolution. At the time of data acquisition, the CMEMS MultiYear (MY) SSH product was available only through 7 June 2023. Thus, we used the MY product for 1 January 1993–7 June 2023 and the NRT product for the remaining period. Notably, the DUACS-processed CMEMS SSH product has since been updated to $0.125^\circ \times 0.125^\circ$ resolution, and the previously used $0.25^\circ \times 0.25^\circ$ version is no longer available. Sea surface wind (SSW) was obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis (ERA5) and represents the zonal (u) and meridional (v) 10 m wind component (Hersbach et al., 2023) at hourly $0.25^\circ \times 0.25^\circ$ resolution. Net heat flux (NHF) was computed from ERA5 as the sum of shortwave radiation, longwave radiation, sensible heat flux, and latent heat flux. Tide, represented by the zonal (u), meridional (v), and vertical (z) components of tidal velocity, was reconstructed from the TPX09 global tide model developed by Oregon State University (OSU) (Egbert and Erofeeva, 2002) using 14 tidal constituents (K_1 , K_2 , M_2 , M_4 , MF , MM , MN_4 , MS_4 , N_2 , O_1 , P_1 , Q_1 , S_1 , and S_2). Depth was

obtained from the General Bathymetric Chart of the Oceans (GEBCO) global terrain model (GEBCO Bathymetric Compilation Group, 2023), which includes the longitude–latitude grid information.

Sea surface temperature (SST) and Sea surface salinity (SSS) were obtained from *in-situ* observations and the CMEMS Global Ocean Physics Reanalysis (GLORYS12V1) (Lellouche et al., 2021). GLORYS12V1 was initially considered as the sole source of SST and SSS. However, comparison with *in-situ* data showed non-negligible errors, with RMSEs of 1.39 °C and 0.84psu. Therefore, *in-situ* data at 0 m depth were used as training input, whereas GLORYS12V1 surface data were used as spatially complete surface inputs to produce 3D Sub_T and Sub_S fields. Here, the *in-situ* data used in SST and SSS are described in Section 2.2.

All sea surface variables were resampled to daily $0.25^\circ \times 0.25^\circ$ resolution. However, hourly variables (Tide, SSW, and NHF) were not daily-averaged to preserve short-term variability; hourly variables were matched to the *in-situ* observation times.

2.2 Subsurface data

All subsurface data were from *in-situ* observations and used as ground truth (Figure 1B). The National Institute of Fisheries Sciences (NIFS) provided subsurface temperature (Sub_T) and subsurface salinity (Sub_S) profiles over 0–500 m depths from conductivity-temperature-depth (CTD) observations (Intergovernmental Oceanographic Commission, 2013). The NIFS dataset has 9,866 profiles, of which approximately 22% (n=2,170) were classified as profiles located within eddies. This classification was based on the Okubo–Weiss parameter (Okubo, 1970), with eddies defined in this study as regions satisfying $OW < -0.2\sigma$.

The NIFS profiles do not fully cover the study area and are sparse during the intervening months between regular bimonthly surveys, accounting for approximately 6% of the number of profiles obtained during the regularly sampled months (Figure 1A). Therefore, Argo profiles archived at the Global Data Assembly Centers (GDACs) by the Korea Institute of Ocean Science and Technology (KIOST) and the Korea Meteorological Administration (KMA) were added to reduce these spatial and temporal gaps (Argo, 2025), providing near-complete spatial coverage and a more uniform monthly distribution.

TABLE 1 Summary of the data used in this study.

Data type	Parameter	Source	Resolution (spatial/temporal)	Purpose
Surface data (input)	SST	CMEMS	$0.08^\circ \times 0.08^\circ$ /daily	Produce 3D fields
	SSS			
	SSH		$0.25^\circ \times 0.25^\circ$ /daily	Training/test set & Produce 3D fields
	SSW _u , v	ECMWF	$0.25^\circ \times 0.25^\circ$ /hourly	
	NHF			
	Tide _u , v, z	TPX09	–	
Subsurface data (output)	Sub _T , S	GEBCO	–	
		NIFS	Station (0–500 m depths)	Training/test set
		KMA		
		KIOST		Independent test set

All sea surface data were resampled to a daily temporal resolution on a $0.25^\circ \times 0.25^\circ$ grid. As an exception, hourly data such as sea surface wind (SSW), net heat flux (NHF), and tide (Tide) were not daily averaged; instead, the variables were matched to the observation times of the subsurface data.

Argo profiles were preprocessed only using the quality control flag, following Ma et al. (2024) and the Argo user's manual (Argo data management, 2022; v3.41.1). Processing steps were: selection of delayed-mode data; quality control using flags 1–2 for pressure, temperature, and salinity (flag 1 for time and position); vertical interpolation to 0–500 m depths for consistency with NIFS; and depthwise outlier removal using a 99% confidence interval. After processing, we have 4,100 KIOST and 3,469 KMA profiles. Of these profiles, 24% of the KIOST profiles ($n=984$) and 28% of the KMA profiles ($n=971$) were classified as being located within eddies.

3 Methods

This study was conducted through three steps (Figure 2).

3.1 Production of subsurface data

Sub_T and Sub_S were produced using a CCM. Following the conditional computation concept of Bengio et al. (2015), the CCM comprises a decision module and two specialized branches (eddy and non-eddy modules) (Figure 2A). The use of separate eddy and non-eddy modules was found to mitigate the effects of sampling imbalance between *in-situ* profiles collected within and outside eddies. The decision module routes samples using the OW parameter derived from input data: $OW < -0.2\sigma$ to the eddy module, and otherwise to the non-eddy module. Here, σ denotes the standard deviation of the OW time series at each grid point over 1993–2023. Each routed sample was then processed by the corresponding eddy or non-eddy module to estimate the Sub_T and Sub_S profiles. Note that the eddy and non-eddy modules shared the same architecture

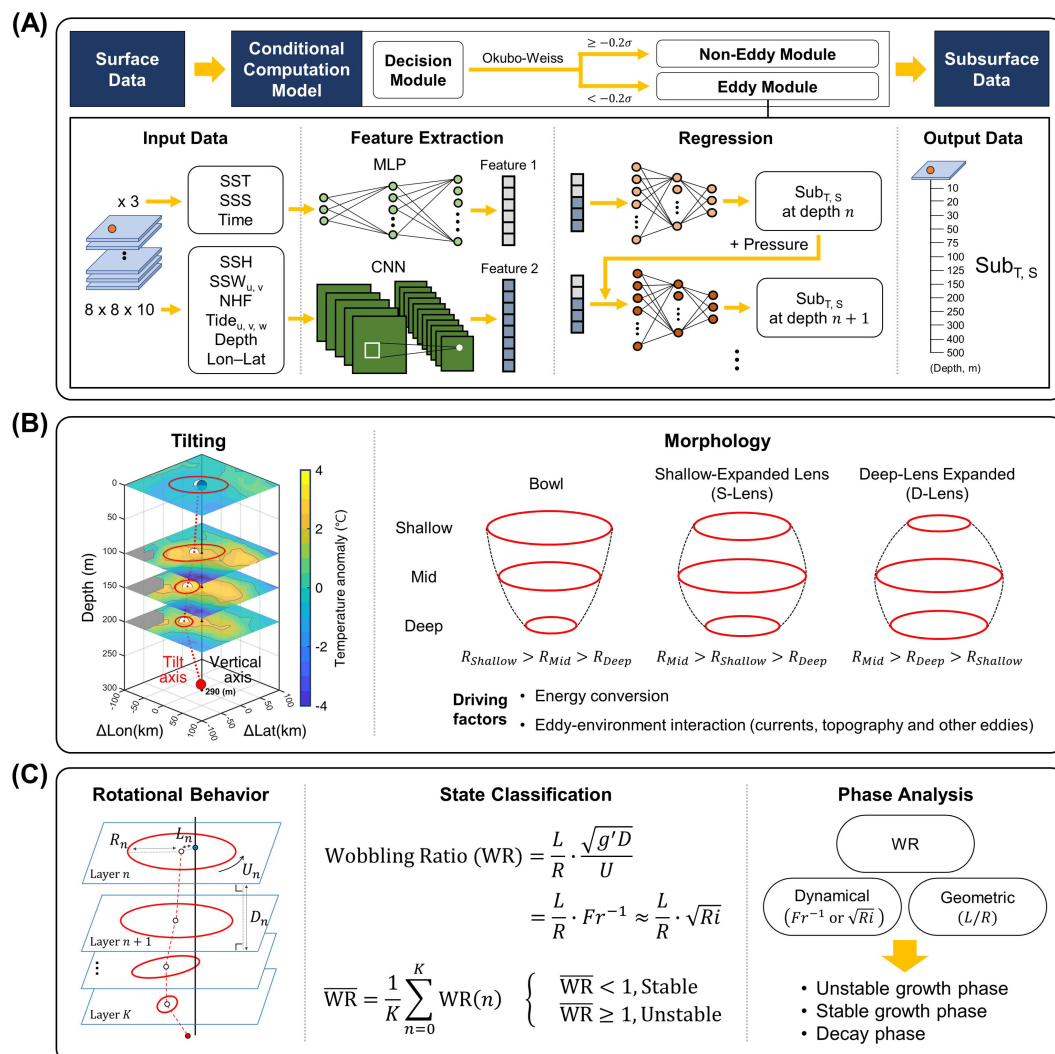


FIGURE 2

Schematic of the study workflow. **(A)** Production of subsurface data. **(B)** Determination of the three-dimensional (3D) structure. The red ellipses delineate the Ulleung Warm Eddy (UWE) boundary; and the white, red, and blue dots indicate the UWE centers, the deepest point of the eddy core, and the location of maximum sea surface height (SSH) within the eddy core, respectively. **(C)** Phase analysis of UWE based on Wobbling Ratio (WR). Variables are defined as follows: L , horizontal offset between the tilt and the vertical axes through the UWE center; R , the mean eddy radius; D , layer thickness; U , boundary-mean velocity; and $g' = g\Delta\rho/\rho_0$, reduced gravity, where ρ_0 is reference density of sea water and $\Delta\rho$ is the density anomaly. The Froude number is defined as $Fr = \sqrt{g'D}/U$, and the Richardson number as $Ri = N^2/(\partial u/\partial z)^2$, where N is the Brunt-Vaisala frequency and $\partial u/\partial z$ is the vertical shear.

for feature extraction and regression, but focal loss was applied with $\gamma = 1.5$ and 2.0, respectively.

The inputs were separated into patch-based and vector-based formats to extract complementary spatial and pointwise features. Patch inputs comprised SSH, SSW_{U-V} , NHF, $Tide_{U-V-Z}$, Depth, and longitude–latitude, whereas vector inputs comprised SST, SSS, and time. Patches were provided as 8×8 grids, and vectors corresponded to the central grid point (4, 4) of each patch. A convolutional neural network (CNN) encoded the patch inputs and a multilayer perceptron (MLP) encoded the vector inputs in feature extraction layer; the resulting features were concatenated and passed to a regression layer. Sub_T and Sub_S were estimated sequentially at 13 levels spanning 10–500 m depths using a residual framework conditioned on temperature, salinity, and pressure from the previous depth, together with the extracted features. Finally, the estimated profiles were assembled into daily 3D fields on a 25-km grid.

For model training and evaluation, all samples were preprocessed by removing cases with more than 50% missing values within the input patch and standardizing all variables, with outputs standardized separately at each depth. The preprocessed NIFS and KMA samples ($n = 13,335$) were shuffled and split, with 80% used for training and 20% for validation. The preprocessed KIOST samples ($n = 4,100$) were used as an independent test set for an objective performance evaluation. Performance was quantified at each depth using the coefficient of determination (R^2), root-mean-square error (RMSE), and normalized RMSE (NRMSE).

3.2 Determining the 3D structure of UWE

UWE's 3D structure was determined using the Lagrangian Particle Tracking (LPT) following Jo et al. (2017). LPT delineates eddy boundaries by identifying particle aggregation within eddy-induced closed streamlines and applying a particle-density threshold, which is set based on the Rossby radius (Kim et al., 2024; see also Arruda et al., 2004).

LPT was applied at the surface over 1993–2023, detecting a total of 59 UWE cases (Figure 3). Among them, the longest-lived case (November 2014–August 2016) was selected, and its 3D structure

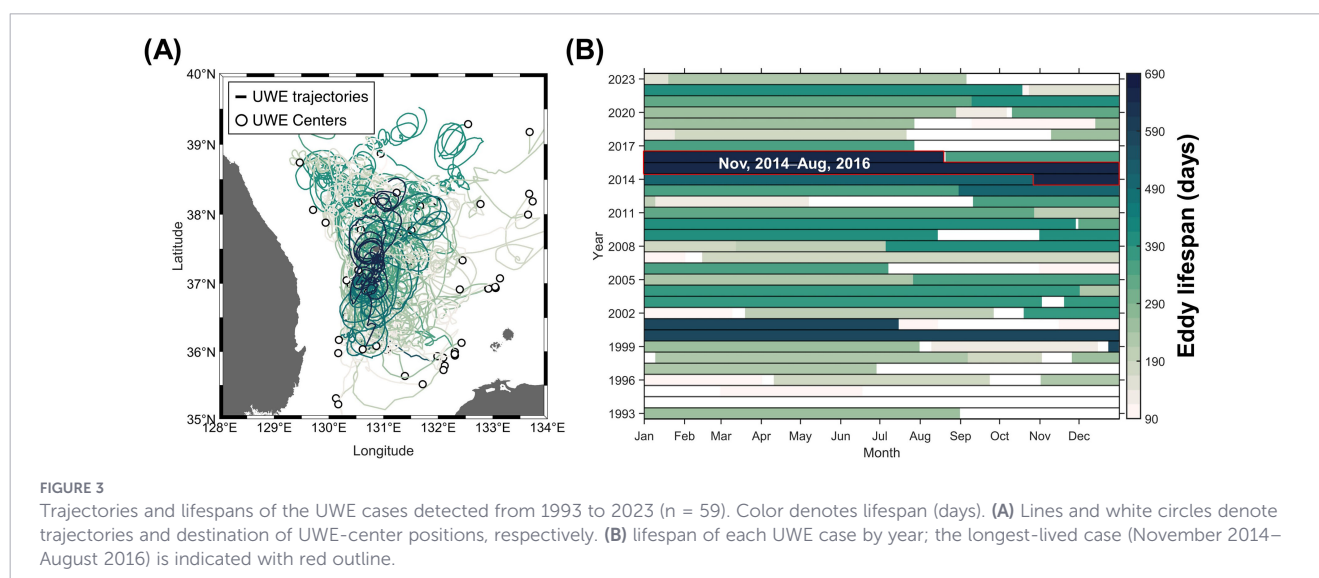
was determined by applying LPT at 0, 100, 150, and 200 m depths, consistent with the reported maximum formation depth (Shin et al., 2019) (Figure 2B). Here, 3D geostrophic velocity field was reconstructed from the CCM-derived Sub_T and Sub_S fields following Kosempa and Chambers (2014), using 500 m as the reference depth.

From the depth-resolved UWE centers and radii, we determined tilting and morphology of 3D structure. Tilting was quantified by the zenith angle θ , defined as the angle between the tilt axis and the vertical axis (Figure 2B). Here, the vertical axis was defined as the vertical line passing through the location of maximum SSH within the eddy, and the tilt axis as the line connecting the UWE centers to the deepest point of the eddy. Morphology was classified from the vertical distribution of eddy radius across three depth ranges: shallow (0 m), mid (100 and 150 m), and deep (200 m). At each depth, the eddy radius was defined as the mean of the semi-major and semi-minor axes of the fitted ellipse. Cases with a maximum radius at mid-depth were classified as lens-shaped and further subdivided into surface-intensified and laterally expanded structure (S-lens) and deep-reaching, laterally extensive eddy core (D-lens), depending on whether the radius in the shallow or deep depth was larger. Cases with a monotonic decrease in radius with depth were classified as bowl-shaped.

Variations in θ and morphology were used as distinct but complementary structural diagnostics to qualitatively infer the dominant processes controlling the three-dimensional structure of the UWE, because θ represents the depth-dependent displacement of the eddy-shape center, whereas morphology describes the vertical variation of the LPT-derived eddy radius. Together, these diagnostics can reflect the response of the UWE structure to surrounding geostrophic flow and eddy-environment interactions involving currents, topography, and neighboring eddies (Kida, 1981; Li et al., 2022; Zhang et al., 2016).

3.3 Phase analysis of UWE

This study aims to analyze the kinematic phases of the UWE using WR. To demonstrate the utility of WR, we conducted phase analyses from two perspectives: energy budget and WR.



3.3.1 Energy budget

In this study, we use the energy balance equation of [Orlanski and Cox \(1973\)](#), following approaches in the previous studies. In this equation, the eddy total energy (ETE) is defined as the sum of perturbation's kinetic energy (EKE) and potential energy (PEP). Note that EKE is often referred to as the eddy kinetic energy. The time tendency of ETE is governed by four contributions: the divergence, baroclinic conversion (BCI), barotropic conversion (BTI), and a Kelvin-Helmholtz instability (KH) term ([Equations 1, 2](#)).

$$\begin{aligned} \frac{d}{dt} \left[\frac{1}{2} (\overline{u'^2} + \overline{v'^2}) + \frac{g^2}{2N^2} \frac{\overline{\rho'^2}}{\rho_0^2} \right] \\ = -\nabla \cdot \left(\frac{\overline{U'p'}}{\rho_0} \right) - \left(\overline{u'^2} \frac{\partial \bar{u}}{\partial x} + \overline{v'^2} \frac{\partial \bar{v}}{\partial y} + \overline{u'v'} \frac{\partial \bar{u}}{\partial y} + \overline{u'v'} \frac{\partial \bar{v}}{\partial x} \right) \\ - \frac{g^2}{\rho_0^2 N^2} \left(\overline{u'\rho'} \frac{\partial \bar{\rho}}{\partial x} + \overline{v'\rho'} \frac{\partial \bar{\rho}}{\partial y} \right) - \left(\overline{w'u'} \frac{\partial \bar{u}}{\partial z} + \overline{w'v'} \frac{\partial \bar{v}}{\partial z} \right) \quad (1) \end{aligned}$$

$$\frac{d}{dt} (\text{EKE} + \text{PEP}) = \text{Divergence} + \text{BTI} + \text{BCI} + \text{KH} \quad (2)$$

Where an overbar denotes the daily climatology, obtained by averaging each day over 1993–2023, and a prime denotes the anomaly relative to this climatology. u , v , and w are the zonal, meridional, and vertical velocity components, with primes denoting anomalies; U' is the velocity vector anomaly; p' and ρ' are the pressure and density anomalies; ρ_0 is a reference density of sea water; g is gravitational acceleration; and N is the Brunt-Vaisala frequency.

To interpret eddy phase from an energetics perspective, we focus on BTI and BCI as the primary conversion pathways between the mean flow and the perturbation field. This focus is justified because BTI and BCI directly represent the barotropic and baroclinic mean-to-eddy energy conversions, respectively, which are the conversion processes most relevant to eddy growth and

decay ([Gill et al., 1974](#); [Li et al., 2020](#)). The divergence and KH terms are not used as conversion diagnostics; the divergence term represents redistribution of eddy energy across the eddy boundary rather than mean-to-eddy conversion ([Oey, 2008](#); [Pope, 2000](#)); and the KH term is also treated as unresolved because it requires vertical Reynolds stresses, $\overline{w'u'}$ and $\overline{w'v'}$, which are not available from the geostrophic diagnostics ([Pedlosky, 1987](#); [Vallis, 2017](#)). Consequently, the energetics-based phase analysis relies on EKE, PEP, BTI, and BCI ([Equation 3](#)).

$$\frac{d}{dt} (\text{EKE} + \text{PEP}) \approx \text{BTI} + \text{BCI} \quad (3)$$

3.3.2 Wobbling Ratio

The Wobbling Ratio (WR) proposed in this study is a dimensionless diagnostic index for quantifying the relative stability of the three-dimensional tilt structure of the UWE. Its formulation starts from the observed tilted rotational behavior of the UWE. As shown in [Figure 4A](#), the UWE center exhibits rotational displacement around the reference vertical axis defined near the SSH maximum ([Walsh et al., 1996](#)). In addition, [Figure 4B](#) shows that the azimuthal current speed increases approximately with radial distance within the inner core, indicating that a representative rotational scale can be defined for the solid body ([Valencia et al., 2025](#)). This solid-body behavior implies that the eddy interior rotates coherently around the tilt axis. Therefore, the observed rotational structure motivates the use of a spinning-top analogy, not as a direct dynamical equivalence, but as a conceptual framework for tilted rotational motion.

The analogy is introduced through the precession–nutation problem of a spinning top, in which the stability of a tilt axis is controlled by the balance between axis displacement and torque. Following [Cross \(2013\)](#), we can describe the motion of a spinning top about the precession axis using the precession–nutation framework ([Figure 5](#)). The corresponding stability condition can be written as.

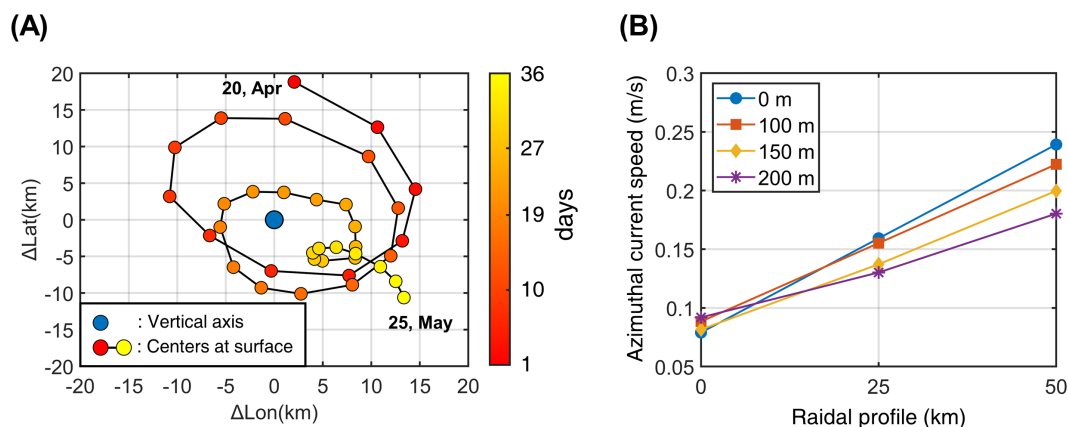


FIGURE 4

Rotational characteristics of the UWE. (A) The blue dot marks the reference vertical axis defined near the maximum SSH within the UWE, and dots colored from red to yellow indicate the surface positions of the UWE center, with color denoting temporal progression in days. (B) Radial profiles of azimuthal current speed relative to the UWE center at each depth, averaged over November 2014 to September 2015; blue, orange, yellow, and purple solid lines denote the 0, 100, 150, and 200 m depths, respectively.

$$\frac{2\sqrt{(I_{cm} \cos \theta + MH_1 H_2)MgH_2}}{I_2 \omega_2} < 1 \quad (4)$$

where I_{cm} is the moment of inertia about the vertical axis through the center of mass G , M is the mass of the top, H_1 is the vertical height of G , H_2 is the distance from the contact point P to G , g is gravitational acceleration, I_2 is the moment of inertia about the spin axis, ω_2 is the spin angular velocity, and θ is the zenith angle of the spin axis (Figure 5A).

Using Equation 5.

$$H_1 = H_2 \cos \theta \quad (5)$$

and Equation 6.

$$I_{cm} = ML^2 - MH_2^2, \quad (6)$$

where L is the distance between the spin and precession axes measured from G , the term inside the square root in Equation 4 becomes Equation 7.

$$\begin{aligned} (I_{cm} \cos \theta + MH_1 H_2)MgH_2 \\ = [(ML^2 - MH_2^2) \cos \theta + M(H_2 \cos \theta)H_2]MgH_2. \end{aligned} \quad (7)$$

The two $MH_2^2 \cos \theta$ terms cancel, yielding Equation 8.

$$(I_{cm} \cos \theta + MH_1 H_2)MgH_2 = M^2 L^2 g H_1. \quad (8)$$

Therefore, Equation 4 reduces to.

$$\frac{2ML\sqrt{gH_1}}{I_2 \omega_2} < 1, \quad (9)$$

Equation 9 represents a stability ratio associated with the tilt axis. When this ratio is less than 1, θ remains nearly constant and the spin axis follows a nearly steady circular orbit. In contrast, when the ratio approaches or exceeds 1, perturbations in θ increase in amplitude, leading to a more pronounced wobbling trajectory (Figure 5B).

Equation 9 is not applied directly to the oceanic eddy. Instead, it is worth to note that all parameters can be interpreted in terms of representative eddy-core scales. This interpretation is consistent with previous descriptions of intrathermocline eddies as lens-shaped, relatively homogeneous rotating cores and with recent observations of a solid-body-like inner core in an intrathermocline eddy (Dugan et al., 1982; Armi and Zenk, 1984; McWilliams, 1988; Chelton et al., 2011; Valencia et al., 2025). To elaborate, we retain only its dimensionless structure and express the individual parameters in terms of corresponding solid-body rotation. In this scaling formulation, M is replaced by the eddy-core mass M_e , L is interpreted as the horizontal offset between the tilt axis and the vertical axis through the UWE center, and the gravitational velocity scale $\sqrt{gH_1}$ is replaced by the reduced-gravity internal-wave speed scale,

$$c = \sqrt{g'D}, \quad (10)$$

where g' is the reduced gravity and D is the effective layer thickness of the coherent eddy core. The spin-axis moment of inertia is approximated as.

$$I_2 \sim M_e R^2, \quad (11)$$

where R is the representative eddy radius. The angular velocity scale is approximated as.

$$\omega_2 \sim \frac{2U}{R}, \quad (12)$$

where U is the boundary-mean azimuthal velocity, following the vorticity in solid-body rotation (Vallis, 2017). This approximation is supported by the solid-body rotation observed in Figure 4B, but it does not imply rigid-body rotation of the entire eddy.

With these substitutions, Equation 9 becomes.

$$\frac{2M_e L \sqrt{g'D}}{(M_e R^2)(2U/R)}, \quad (13)$$

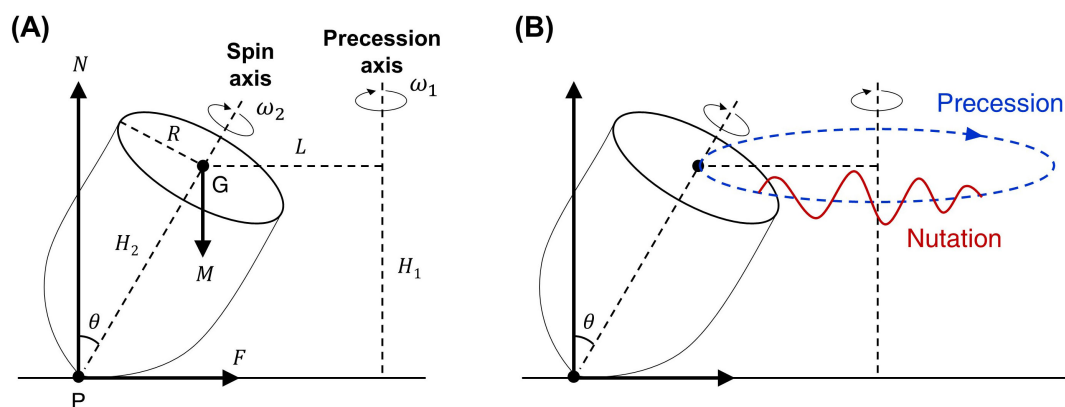


FIGURE 5

Schematic of spinning-top motion about the precession axis in the precession–nutation framework. (A) ω_1 and ω_2 denote the angular velocities of the spin and precession axes, respectively. Here, L denotes the distance between the spin and precession axes measured from the top's center of mass G ; R denotes the radius of the top; H_1 denotes the height of G above the floor; H_2 denotes the distance from the contact point P to G ; θ denotes the zenith angle of the spin axis; and M denotes the mass of the top. The vertical reaction force N and the centripetal force F are also indicated. (B) The blue dashed and red solid lines denote the precession and nutation trajectories, respectively. The spinning-top analogy is motivated by solid-body-like rotation within coherent eddy core, but it is not applied directly on oceanic eddy.

which reduces to.

$$WR = \frac{L}{R} \cdot \frac{\sqrt{g'D}}{U}. \quad (14)$$

Defining the internal Froude number as.

$$Fr^{-1} = \frac{U}{c} = \frac{U}{\sqrt{g'D}}, \quad (15)$$

WR can be written as.

$$WR = \frac{L}{R} \cdot Fr^{-1}. \quad (16)$$

Thus, WR combines a geometric component, L/R , and a dynamical component, Fr^{-1} . The geometric term represents the normalized displacement of the tilt axis, whereas the inverse Froude number represents the reduced-gravity internal-wave speed scale relative to the azimuthal velocity scale of the coherent eddy core. Under the additional approximations $c = \sqrt{g'D} \approx ND$ and $\partial u / \partial z \approx U/D$, Fr^{-1} can also be interpreted approximately as $\sqrt{Ri}$, where Ri is the Richardson number.

In this study, the $O(1)$ scale of WR is used as a practical reference for distinguishing relatively stable and unstable tilted-rotation states. Specifically, $WR < 1$ indicates a relatively stable state in which the tilt remains within an allowable range, whereas $WR \geq 1$ indicates an unstable state associated with wobbling rotation. Over the eddy lifespan, WR exhibited a characteristic U-shaped evolution, with values greater than 1 in the initial stage, less than 1 during the intermediate stage, and greater than 1 again in the late stage. On the basis of this pattern, the kinematic phases were classified as unstable growth, stable growth, and decay. WR is therefore used as an analogy-motivated, physics-based structural diagnostic that complements the energy-budget analysis of UWE phase evolution.

3.4 Experiment setup

We partitioned the selected period (November 2014–August 2016) into March–October 2015 and January–May 2016, denoted as UWE15 and UWE16, respectively. These intervals were selected because the LPT could robustly determine the 3D structure of the UWE during these periods.

To enable WR-based phase analysis, we identified regimes and shifts in the time series of WR using the Sequential t-test Analysis of Regime Shifts (STARS) (Rodionov, 2004). Here, a regime was defined as an interval over which the time series exhibits approximately stationary statistical characteristics, quantified by the interval mean, and shifts were defined as statistically significant step changes separating consecutive regime means. The window length, which is the cut-off length for regime identification and shift detection, was set to 60 days to represent the dominant kinematic time scale of UWE, as indicated by a common peak in fast Fourier transform (FFT) spectra of sea level anomaly (SLA), EKE, and eddy radius (Samelson et al., 2014; Zhang et al., 2022). Each detected WR regime was interpreted as a kinematic phase (unstable growth, stable growth, or decay) and was analyzed in conjunction with energy budget.

4 Results

4.1 Model performance in reconstructing subsurface fields

Figure 6 validates the CCM-produced Sub_T and Sub_S fields using the independent test set. Figure 6A shows the spatial distribution of the full independent test set and a subset of the independent test set within the Ulleung Basin. Because the ARGO-derived independent test set is not spatially uniform, it is matched with the corresponding CCM grid.

Figures 6B, C compare the depth-dependent RMSE profiles. For Sub_T , the test set and the full independent test set are quantitatively comparable, with a maximum difference of 0.19 °C across depths, corresponding to an 11% difference when normalized by the latter. RMSE is 1.13 °C in the test set and 1.30 °C in the independent test set at 10 m depth, increased to maxima of 2.17 °C and 2.28 °C, respectively, at 100 m depth. The depth-averaged Sub_T RMSE is 1.23 °C for the test set and 1.30 °C for the full independent test set, which is slightly lower than the 1.33 °C reported by Lee et al. (2026). However, this comparison should be interpreted cautiously because the validation domain, spatial resolution, and sampling strategy differ between studies.

For Sub_S , RMSE is 0.16psu in the test set and 0.12psu in the full independent test set at 10 m depth, with a maximum difference of 0.03psu (28%) across depths. The maximum RMSE reach 0.19psu in the test set and 0.15psu in the independent test set at 20 m depth, and then decrease with depth. Within the Ulleung Basin subset, RMSE remain at or below 1.62 °C for Sub_T and 0.07psu for Sub_S across all depths. These lower RMSE values indicate that the CCM errors were relatively small within the Ulleung Basin, where the UWE forms, compared with the full domain.

Figures 6D, E show the horizontal distributions of RMSE at selected depths from 10 to 300 m. Sub_T RMSE exhibit relatively high errors northwestern domain at 10 m depth and along the near-coastal boundary at 50–150 m depth, whereas errors generally decrease at depths of 200–300 m. In addition, Sub_S RMSE also show relatively large errors mainly confined to the northwestern domain and reduced errors below 50 m depth.

Figures 6F–I further compare the observed and CCM-produced time–depth profiles from the independent test set. The CCM generally reproduce the seasonal evolution of the vertical structures. However, the reconstructed profiles are smoother than the observations, particularly in some profiles above 200 m depth.

4.2 Assessment of the UWE 3D structure

Figure 7 shows time series of depth-dependent radius and tilt index, together with representative snapshots of the three-dimensional structure of UWE15 from March to October 2015. Here, overbar denotes the mean of θ evaluated at 100, 150, and 200 m depth, as well as at the depth of the deepest point of the eddy. During March–May, UWE15 evolved from an S-lens structure to a D-lens structure (Figures 7A, B). On 3 April, UWE15 exhibited an S-lens structure, characterized by a relatively broad surface expression and smaller radii below 100 m, while the tilt parameter

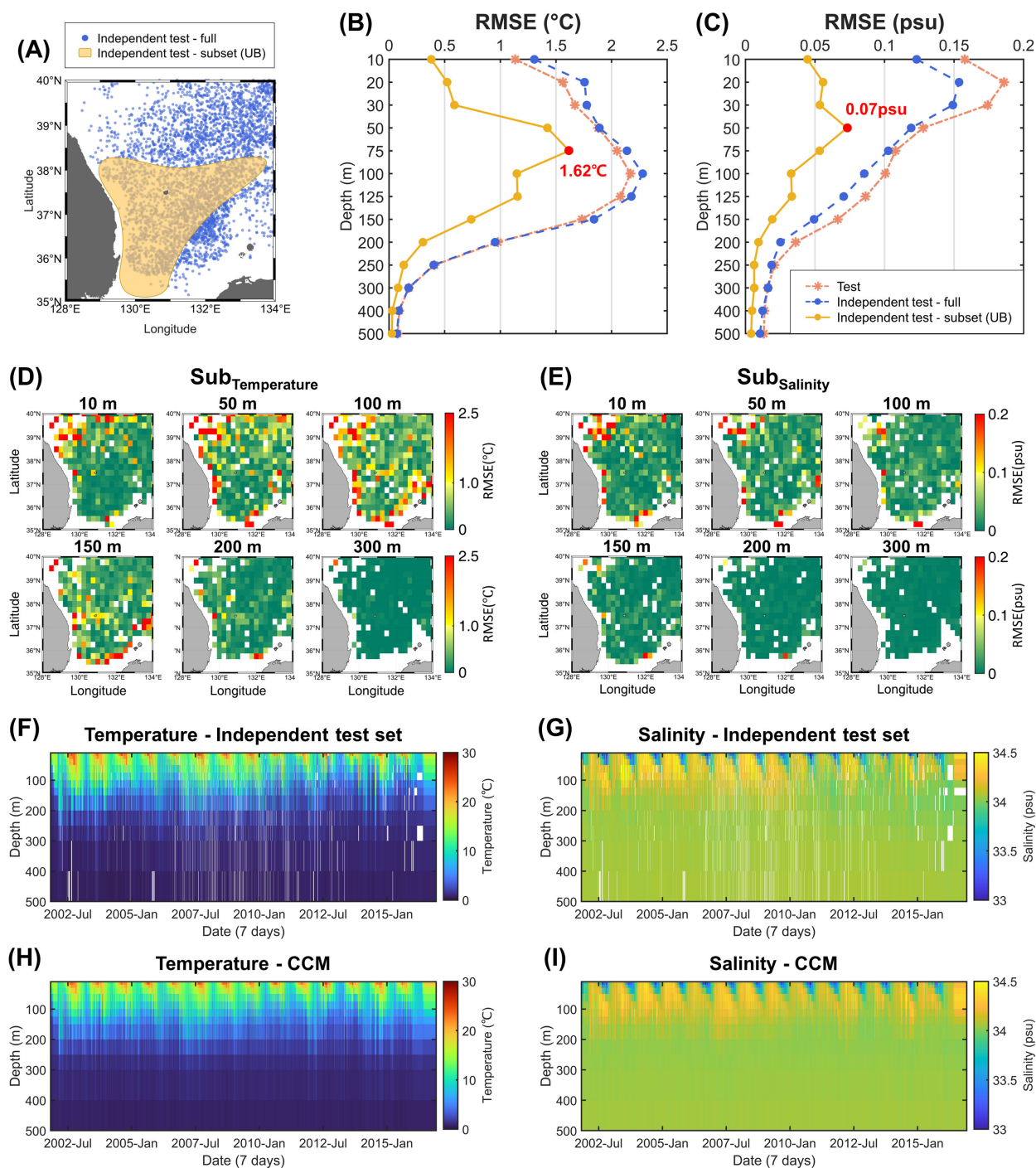


FIGURE 6

Independent validation of CCM-produced subsurface temperature (Sub_T) and salinity (Sub_S). (A) Spatial distribution of the full independent test set, shown as blue dots, with the yellow shaded area denoting the Ulleung Basin subset. (B, C) depth-dependent RMSE profiles for temperature and salinity. Red dashed lines with star markers represent the test set, blue dashed lines with circle markers represent the full independent test set, and yellow solid lines with circle markers represent a subset of the independent test set within the Ulleung Basin. (D, E) Horizontal RMSE distributions at selected depths from 10 to 300 m. (F, G) Observed time–depth distributions from the full independent test set. (H, I) Corresponding CCM-produced profiles.

remained low ($\cos \bar{\theta} = 0.0056$) (Figure 7A). By 18 May, radii had increased at all sampled depths, with pronounced expansion at 100–200 m, and the eddy had evolved into a D-lens configuration accompanied by a larger tilt parameter ($\cos \bar{\theta} = 0.0345$) (Figure 7B). During June–July, the radius continued to increase and reached a maximum in early July, with values of approximately 55–65 km depending on depth (Figure 7E). After July, UWE15

transitioned toward a bowl-shaped structure. On 6 August, the vertical structure exhibited a bowl-shaped morphology, and the tilt parameter had decreased to 0.0066 (Figure 7C). From August onward, radius contraction became pronounced at 150 and 200 m, whereas the boundaries at 0 and 100 m remained comparatively coherent at approximately 45– (Figure 7E). By 30 September, the bowl-shaped structure persisted, and the deep boundary

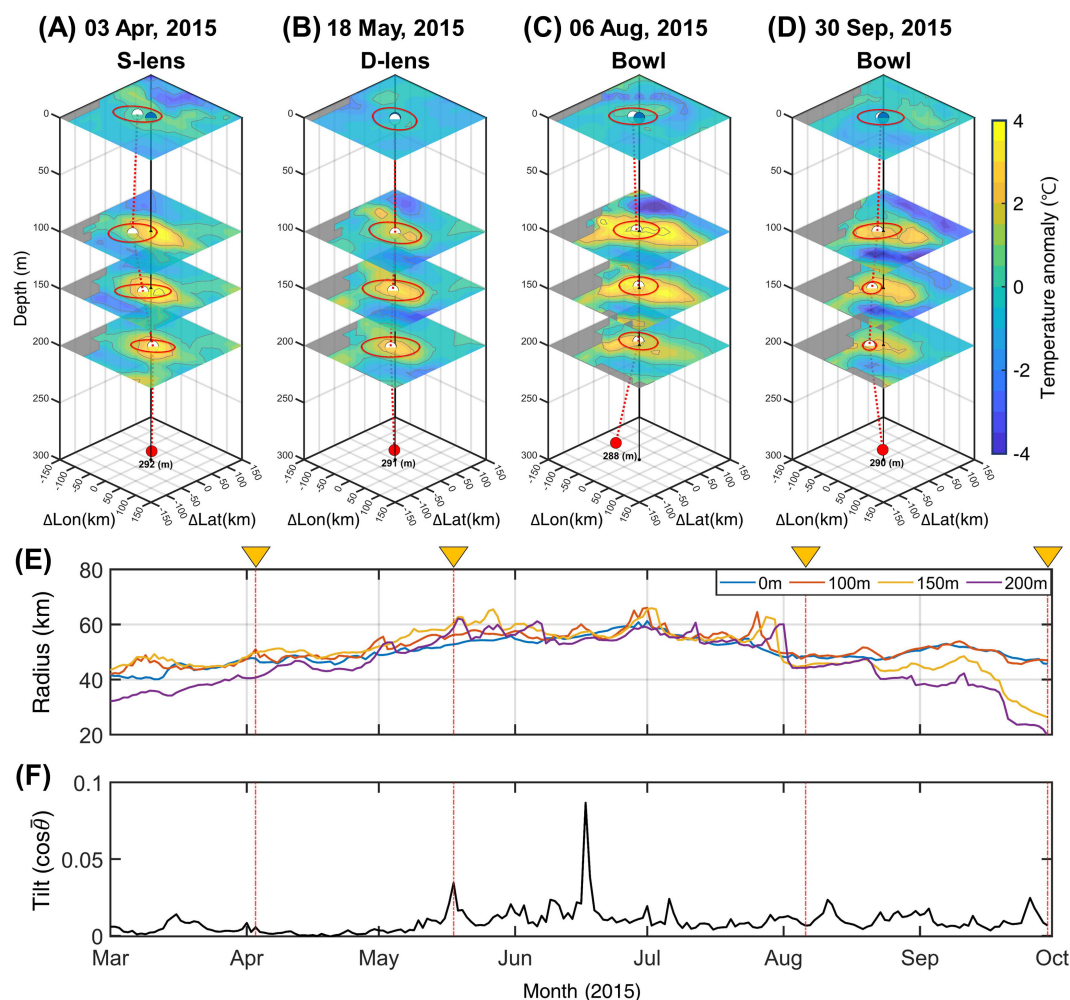


FIGURE 7

Time series of depth-dependent radius and tilt index, together with representative snapshots of the 3D structure of UWE15, where UWE15 denotes the UWE during March–October 2015. UWE15 (A–D) Representative snapshots on 3 April, 18 May, 6 August, and 30 September 2015. The red dashed and black solid lines denote the tilt axis and vertical axis, respectively, and the shading indicates temperature anomaly. Morphological classes are defined in Figure 2B. (E) Eddy radius at 0 m (blue), 100 m (orange), 150 m (yellow), and 200 m (purple); yellow triangles mark the dates shown in (A–D). (F) θ denotes the zenith angle between the tilt axis and vertical axis, and the tilt parameter $\cos \theta$ is defined as the mean of θ evaluated at 100, 150, and 200 m depths, and at the depth of the deepest point of the eddy core.

weakened 50 km (markedly, preceding near-complete boundary loss at 200 m and substantial contraction at 150 m near October (Figures 7D, E). Figure 7F shows that the tilt parameter remained small through most of the record, although brief excursions occurred in mid-May and mid-June, with the largest peak appearing in mid-June.

The morphological sequence from the S-lens to the D-lens and bowl-shaped structures was qualitatively consistent with seasonal variation by the background circulation. Previous studies reported that EKWC transport was minimal in January and maximal in October (Shin et al., 2022). Thus, the progressive radius expansion from spring to early summer and the persistence of a coherent shallow boundary into late summer may be qualitatively compatible with a strengthening background current. However, because EKWC transport was not directly calculated in this study, the late-season transition toward a bowl-shaped structure should be interpreted only as a qualitative hypothesis. One possible explanation is that seasonal variability of the EKWC modified the surrounding shear

and strain field around the UWE, maintaining a broader shallow boundary while deforming or weakening the deeper boundary. This interpretation is consistent with previous studies showing that bowl-shaped eddies can be related to surface-flow instability or wind-stress-curl forcing, and that current–eddy interactions and mean-flow strain can contribute to eddy deformation (Lin et al., 2015; Zhang et al., 2017; Qiao et al., 2023).

4.3 Phase evolution with the energy budget and Wobbling Ratio

Figure 8 shows the evolution of the UWE15's volume (V) together with WR and energy budget terms (EKE, PEP, BCI, and BTI). In this study, V is used as a reference variable to indirectly identify the eddy phases; it is calculated by multiplying the layer thickness by the eddy area. These components are related to eddy radius and amplitude, which are used to diagnose eddy phase evolution (Chelton et al., 2011; Zhang et al., 2022).

In the energy-budget perspective, V increased after March ($dV/dt = 0.0097 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), as both EKE and PEP gradually increased. After entering Event 1 around 18 April, the growth rate of V increased markedly ($dV/dt = 0.0318 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$); during this period, EKE and PEP continued to increase, while BCI and BTI exhibited pronounced negative and positive peaks, respectively. After ETE reached its maximum on 13 May, EKE and PEP showed a decreasing tendency, and V became nearly constant from around 26 May ($dV/dt = 0.0006 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$); it subsequently began to decrease from around 1 July ($dV/dt = -0.0163 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), and this decline continued into Event 2. During Event 2, EKE and PEP showed a transient increase followed by a gradual decrease, while near the ETE maximum, BCI and BTI switched to negative and positive peaks, respectively. Finally, after the end of Event 2, the rate of V decrease further intensified ($dV/dt = -0.0468 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$).

From the WR perspective, the time series of WR was divided into four regimes (P1–P4), each represented by an interval mean WR ($\overline{\text{WR}}$). During P1 ($\overline{\text{WR}} = 1.44$), V increased ($dV/dt = 0.0097 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), and the shift from P1 to P2 occurred after the onset of the rapid increase in V . In the early stage of P2 ($\overline{\text{WR}} = 0.74$), V grew rapidly ($dV/dt = 0.0318 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), reached a local maximum on 26 May, and became nearly constant ($dV/dt = 0.0006 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$). Following the transition to P3 ($\overline{\text{WR}} = 1.13$), V reached another local maximum on 1 July and then gradually decreased ($dV/dt = -0.0163 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$). After the

transition to P4 ($\overline{\text{WR}} = 2.11$), the V decrease intensified relative to P3 ($dV/dt = -0.0468 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$). Overall, based on WR-defined states and the associated volume evolution, UWE15 followed a three-phase sequence of unstable growth, stable growth and decay, with $\overline{\text{WR}}$ exhibiting a U-shaped evolution from 1.44 to 0.74 and then to 2.11.

4.4 Quantitative relationships of eddy volume with the energy budget and WR

Figure 9 shows the correlations of eddy volume with WR and energy-budget during UWE15. Among the relationships, WR shows the largest absolute correlation with eddy volume, with a significant negative correlation ($r = -0.57$, $p = 1.03 \cdot 10^{-19}$). By contrast, $d\text{EKE}/dt$ and $d\text{PEP}/dt$ show weak correlations with eddy volume, with $r = -0.04$ ($p = 0.60$) and $r = -0.05$ ($p = 0.51$), respectively. BCI shows a positive correlation with eddy volume ($r = 0.26$, $p = 9.66 \cdot 10^{-5}$), whereas BTI shows a negative correlation with eddy volume ($r = -0.36$, $p = 0.52 \cdot 10^{-8}$).

The significant negative correlation between WR and eddy volume indicates that higher WR values were generally associated with contracted volume, whereas lower WR values were associated with expanded volume. This relationship is consistent with the conceptual framework of WR. Therefore, among the diagnostics examined here, WR provides the clearest quantitative relationship with the phase evolution of UWE15.

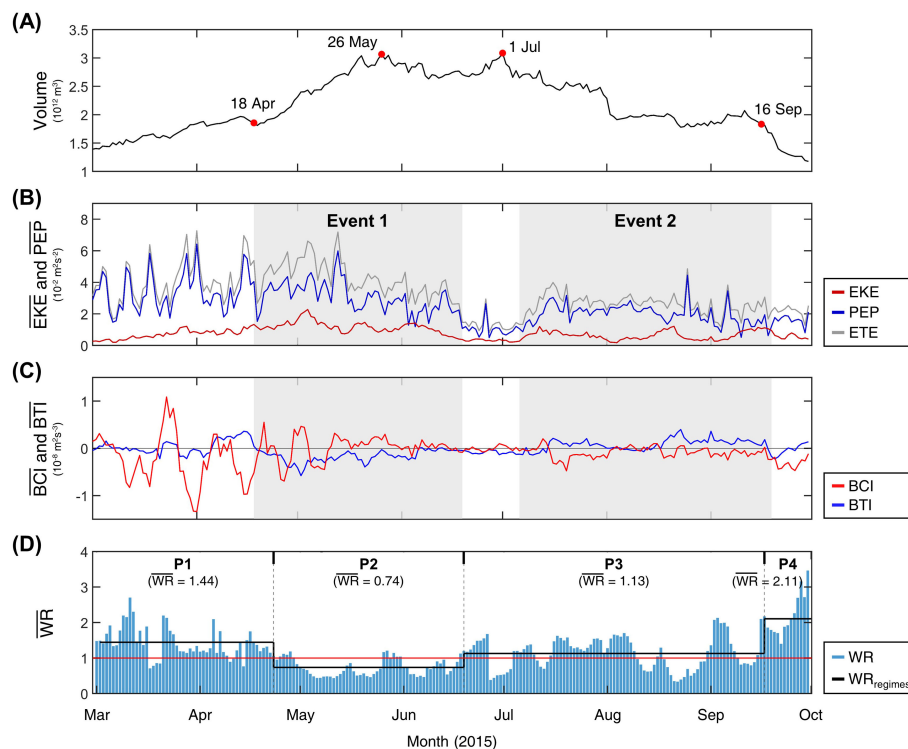


FIGURE 8

Evolution of UWE15 volume, energy-budget terms, and WR. Shaded intervals (Event 1 and 2) highlight periods characterized by prominent peaks in the baroclinic conversion (BCI) and the barotropic conversion (BTI). (A) Red dots denote the reference time points used to estimate the volume tendency. (B) Red, blue and grey lines denote the eddy kinetic energy (EKE), the potential energy of the perturbation (PEP), and the eddy total energy (ETE), respectively; ETE is defined as the sum of EKE and PEP. (C) Red and blue lines denote BCI and BTI, respectively. (D) Blue bars denote WR, and the black step line denotes the mean WR ($\overline{\text{WR}}$) for each interval using the Sequential t-test Analysis of Regime Shifts (STARS).

5 Discussion

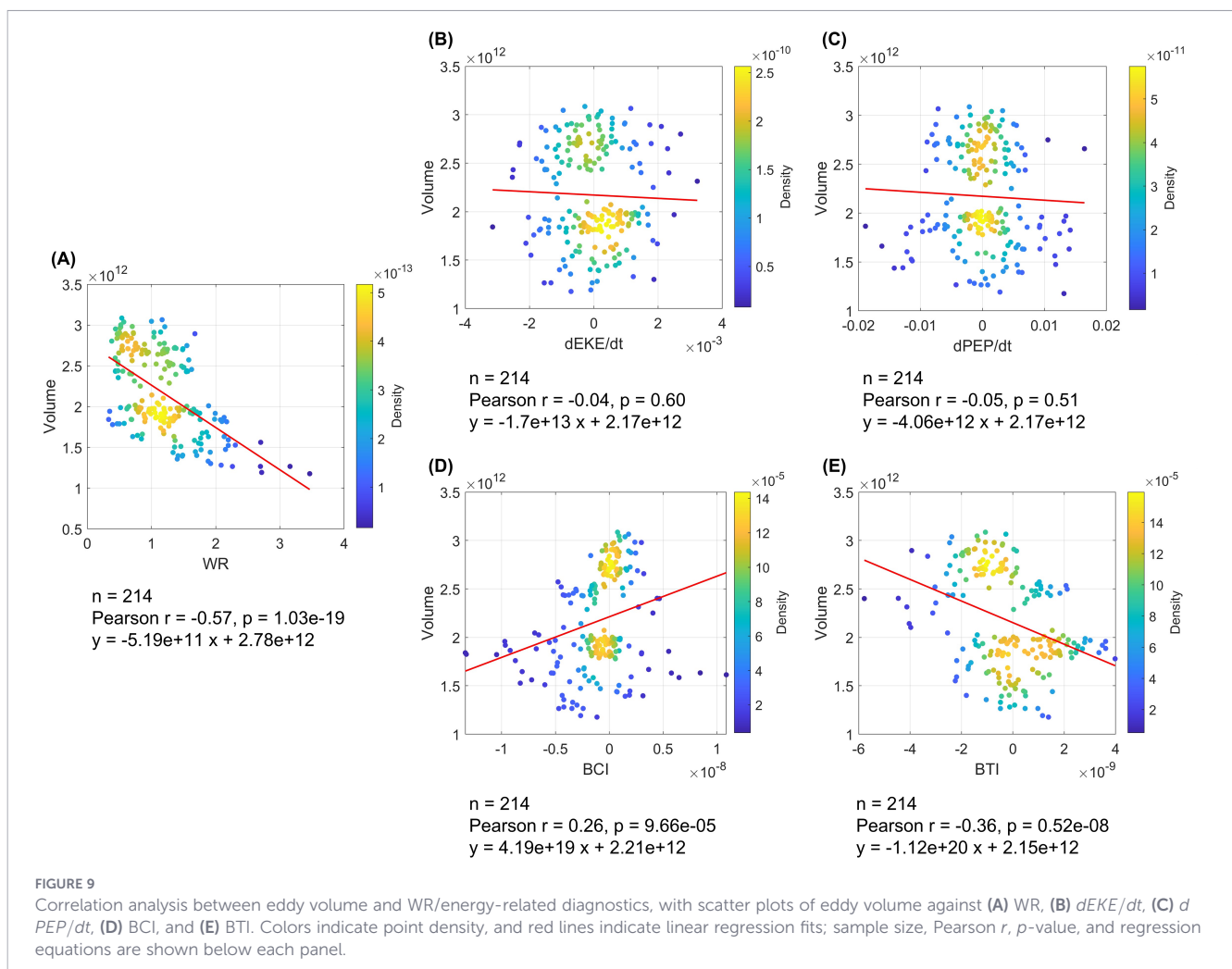
5.1 Physical interpretation of the U-shaped evolution of Wobbling Ratio

Figure 8 showed that WR followed a characteristic U-shaped evolution that corresponded to the unstable-growth, stable-growth, and decay phases identified from P1 to P4. A remaining question is how the U-shaped evolution of WR can be physically interpreted and why WR exhibits that pattern across the kinematic phases. Figures 10A–C provides a conceptual framework for addressing that question by illustrating the spinning-top motion that motivated WR and by relating the geometric and dynamical components of WR to that motion.

Figures 10A–C illustrate three stages of the spinning-top motion. During the initial stage (Figure 10A), the top precesses around the precession axis along a broad orbit that gradually contracts inward, indicating an unstable state, while strong rotational forcing maintains an approximately constant θ . During the intermediate stage (Figure 10B), rotational speed decreases, the orbit contracts toward zero, and the top approaches a nearly stationary position, indicating a stable state. During the late stage (Figure 10C), rotational forcing weakens, the constraint on θ

relaxes, and pronounced wobbling develops, indicating a return to instability.

To evaluate whether the staged spinning-top motion was also expressed in the eddy, the geometric and dynamical components of WR were examined in Figure 10D. The geometric component, L/R , represents the relative displacement from the vertical axis, whereas the dynamical component, $Fr^{-1} = \sqrt{g'D}/U$, reflects the relative strength of the UWE boundary current that, within the spinning-top analogy, corresponds to the rotational forcing. In P1 ($\overline{WR} = 1.14$), the geometric component occupied a relatively high level within its own phase progression, whereas the dynamical component remained comparatively low, indicating that the increase in WR during P1 was more strongly associated with the geometric component. Such behavior is consistent with the initial stage (Figure 10A), in which the top precesses along a broad orbit under strong rotational forcing. In P2 ($\overline{WR} = 0.74$), the geometric component decreased while the dynamical component increased, and WR accordingly declined as the two components approached a more balanced state, consistent with the stable stage shown in Figure 10B, where the orbit contracts toward zero and the top becomes nearly stationary. In P3 ($\overline{WR} = 1.13$), the dynamical component remains relatively high, but the geometric component begins to increase again, indicating a return to an unstable state. In



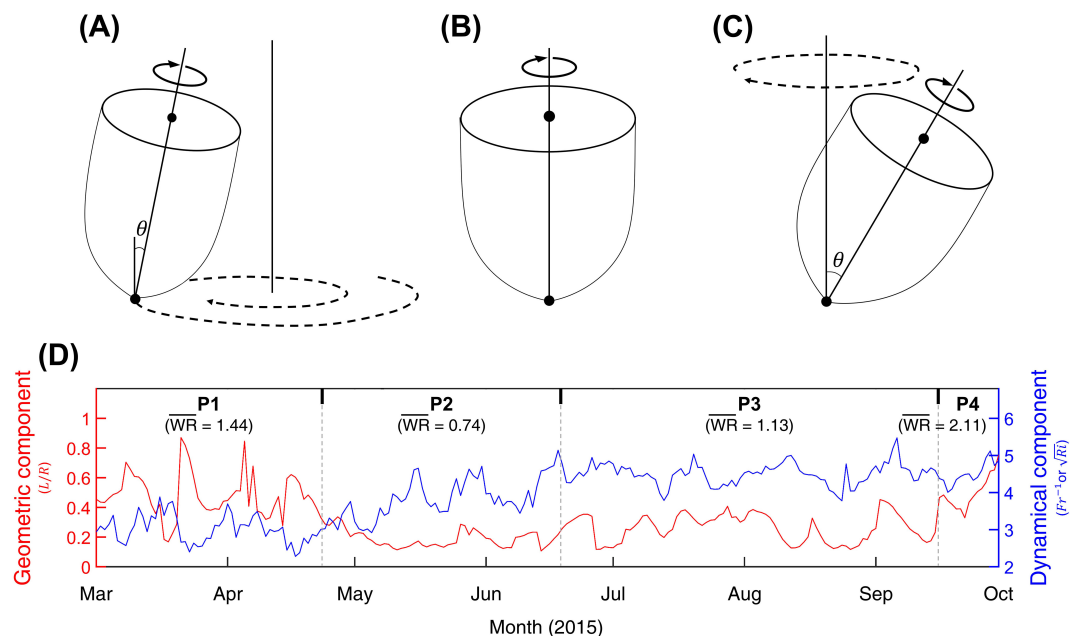


FIGURE 10

Conceptual framework for interpreting the U-shaped evolution of the WR. (A–C) Idealized spinning-top motions corresponding to the initial unstable stage, intermediate stable stage, and late unstable stage, respectively: broad precession with a contracting orbit under nearly constant θ ; near-stationary rotation as the orbit contracts toward zero; and pronounced wobbling accompanied by continued variation in θ . The schematic illustrates how stage-dependent changes in the balance between the geometric and dynamical components produce the U-shaped WR evolution and support phase separation of the eddy life cycle into unstable growth, stable growth, and decay. (D) Time series of the geometric component (L/R) and dynamical component ($Fr^{-1} = \sqrt{Ri}$) of WR during 2015. Upper brackets delineate phases P1–P4, and values beneath the phase labels indicate mean WR for each phase.

P4 ($\overline{WR} = 2.11$), the geometric component increased further, consistent with the late stage shown in Figure 10C, where weakened rotational forcing no longer maintains a nearly constant θ and pronounced wobbling develops.

The U-shaped evolution of WR therefore represents more than an empirical pattern; the evolution reflects stage-dependent changes in the balance between the geometric and dynamical components. WR therefore serves not only as an indicator of stable and unstable states, but also as a physically interpretable index for separating the UWE lifespan into an unstable-growth phase, a stable-growth phase, and a decay phase.

5.2 A case study of divergent phase interpretations

Figure 11 shows the same set of analysis as in Figure 8 but for UWE16. From the energy-budget perspective (FigureS 11B, C), V increased after January ($dV/dt = 0.0137 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), whereas EKE and PEP remained nearly unchanged. Around 9 February, the growth rate of V increased ($dV/dt = 0.0192 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), and by mid-February, PEP increased while both BCI and BTI exhibited positive peaks. V and ETE reached local maxima on 10 March, and V subsequently became nearly constant ($dV/dt = 0.0087 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$). During Event 3, PEP increased further, and BCI and BTI showed larger positive peaks than in March; however, V decreased during the same period ($dV/dt = -0.0193 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$). Such behavior became particularly evident during Event 3, when energetic terms increased despite a decline in V .

The WR-based diagnosis provided a more coherent interpretation of the transition (Figure 11D). The WR time series was partitioned into three regimes (P5–P7). During P5 ($\overline{WR} = 1.43$), V increased ($dV/dt = 0.0137 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$). After the shift to P6 ($\overline{WR} = 0.72$), V increased more rapidly ($dV/dt = 0.0192 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), reached a local maximum on 10 March, and then remained nearly constant ($dV/dt = 0.0087 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$). V reached another local maximum on 22 March and subsequently decreased ($dV/dt = -0.0193 \cdot 10^{12} \text{ m}^3 \text{ day}^{-1}$), coincident with the transition from P6 to P7 ($\overline{WR} = 1.01$). Accordingly, $\overline{WR}$ decreased from 1.43 to 0.72 and then increased to 1.01, forming a U-shaped evolution that is broadly consistent with unstable growth, stable growth, and decay.

A notable discrepancy emerged during P7: V decreased whereas EKE, PEP, BCI, and BTI increased. In contrast, WR increased during P7, consistent with entry into the upward branch of the U-shaped evolution associated with decay. One possible explanation is that the energetic amplification in Event 3 reflected not only the evolution of UWE16 but also interaction with a nearby cyclonic eddy. On 25 May 2016, a cold eddy was located adjacent to and beneath UWE16, forming a vertically connected anticyclonic-cyclonic paired structure (Figure 11E). Because the CE formed near Dok Island, the cyclonic feature is referred to here as the Dok Cold Eddy (DCE). The UWE16-DCE pair may have sharpened the baroclinic slope across the interface between the two eddies, producing downward-concave and upward-convex structures and thereby increasing PEP, as described for interacting eddy pairs by Hunt et al. (2010). Enhanced BCI indicates increased conversion from mean available potential energy to eddy available potential

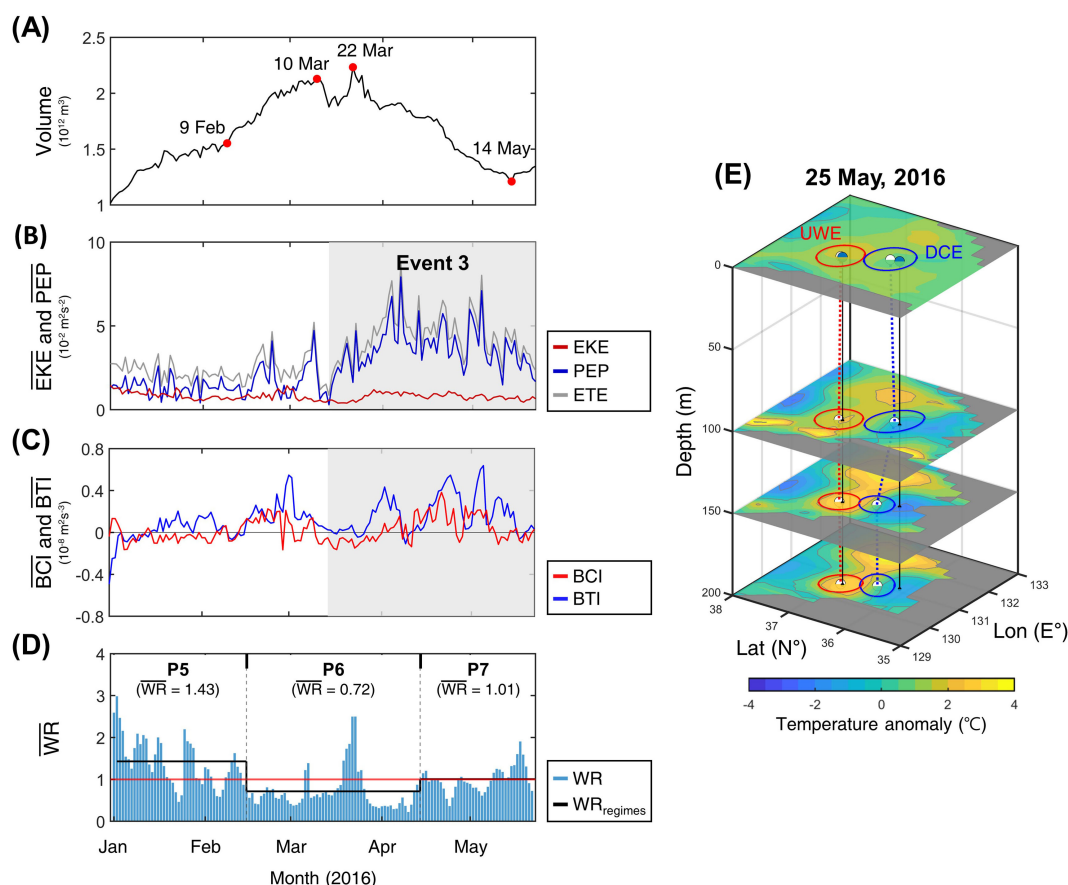


FIGURE 11

Time series of UWE16 (A) volume, (B, C) energy-budget terms, and (D) WR, where UWE16 denotes the Ulleung Warm Eddy during January–May 2016. Details not described here are the same as in Figure 8. (E) Paired 3D structure of UWE16 and the Dok Cold Eddy (DCE) on 25 May 2016. Red and blue ellipses outline the boundaries of UWE16 and DCE, respectively. White dots indicate the eddy centers, and blue dots indicate the local maximum and minimum of absolute dynamic topography (ADT) within each eddy. Dashed and solid lines indicate the tilt and vertical axes, respectively.

energy, represented by PEP in this study, rather than a pathway from PEP to EKE (Gill et al., 1974; Cronin, 1996; Guo et al., 2022; Sun et al., 2022). As the interaction progressed, vorticity annihilation may also have weakened the dipolar structure while making the flow more vertically coherent and horizontally sheared, which could have transiently intensified BTI. Under such conditions, simultaneous increases in PEP, BCI, BTI, and EKE may occur even while UWE16 loses V .

The results suggest that eddy-eddy interaction may have contributed to energetic behavior that was not fully consistent with the phase evolution inferred from UWE16. Such behavior implies that, during Event 3, interpretation based solely on the energy-budget perspective was not straightforward and that consideration of the 3D eddy structure may be helpful in phase analysis. In this context, WR increased from 0.72 to 1.01 across the transition from P6 to P7, consistent with entry into the decay phase (Figure 11D). Taken together, the comparison suggests that WR can serve as a practical complementary index for diagnosing UWE phases, particularly when interpretation based on the energy-budget terms becomes ambiguous.

6 Conclusions

This study proposed the WR, a quantitative index derived from the 3D tilt structure of the UWE, to analyze phases of the UWE. To achieve this goal, the study proceeded in three steps, and the main findings from each step are summarized as follows.

First, the conditional computation model (CCM) produced Sub_T and Sub_S fields from surface data with relatively low errors in the Ulleung Basin. In the subset of independent test within the Ulleung Basin, RMSE remained below 1.62°C for Sub_T and 0.07 psu for Sub_S across all depths, indicating that the CCM provided sufficiently stable subsurface fields for subsequent 3D eddy analysis.

Second, the reconstructed 3D structure of UWE15 showed a sequential morphological transition from an S-lens structure to a D-lens structure and eventually to a bowl-shaped structure. This morphological evolution was qualitatively consistent with the reported seasonal variability of East Korea Warm Current (EKWC) transport.

Third, using these 3D structures, we compared the energy-budget and WR in relation to eddy volume. During UWE15, WR exhibited a U-shaped evolution, decreasing from P1 to P2 in terms of mean WR

(1.44 to 0.74) and then increasing through P3 and P4 (1.13 and 2.11, respectively). This evolution corresponded to a three-phase sequence in which the UWE transitioned from unstable growth to stable growth and finally to decay. The volume-based correlation analysis further showed that, among the energy-budget and WR, WR had the largest absolute correlation with eddy volume, with a significant negative correlation ($r = -0.57$, $p = 1.03 \times 10^{-19}$). This result indicates that lower WR values were generally associated with expanded eddy volume, whereas higher WR values were associated with contracted volume. Therefore, WR provides the clearest quantitative relationship with the volume-based phase evolution of UWE15 among the diagnostics examined in this study.

The UWE16 case demonstrated why a structural diagnostic such as WR is useful as a complement to the energy-budget framework. During P7, eddy volume decreased despite increases in EKE, PEP, BCI, and BTI. This discrepancy is interpreted as a possible consequence of 3D interaction between UWE16 and the neighboring Dok Cold Eddy (DCE), which may have transiently amplified energetic terms while the UWE volume continued to contract. In this case, the energy-budget diagnostics alone did not provide a straightforward interpretation of phase evolution, partly because unresolved nonlinear advection, weak vertical terms, and eddy–eddy interaction can complicate energy-budget closure and the interpretation of energetic tendencies. In contrast, WR increased during P7, consistent with the upward branch of the U-shaped evolution associated with decay. Overall, these findings demonstrate that WR, expressed as $WR = (L/R) \cdot Fr^{-1} \approx (L/R) \cdot \sqrt{Ri}$, is a quantitative index for describing the phase evolution of the UWE. Although WR is not intended to replace energy-budget analysis, it may serve as a complementary 3D structural diagnostic for supporting UWE phase interpretation, particularly when energy-budget tendencies are difficult to interpret.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Author contributions

D-YK: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. EK: Data curation, Writing – review & editing. Y-SJ: Data curation, Methodology, Writing – review & editing. JS: Formal analysis, Validation, Visualization, Writing – review & editing. WL: Formal analysis, Validation, Writing – review & editing. Y-HJ: Formal analysis, Funding acquisition, Supervision, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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